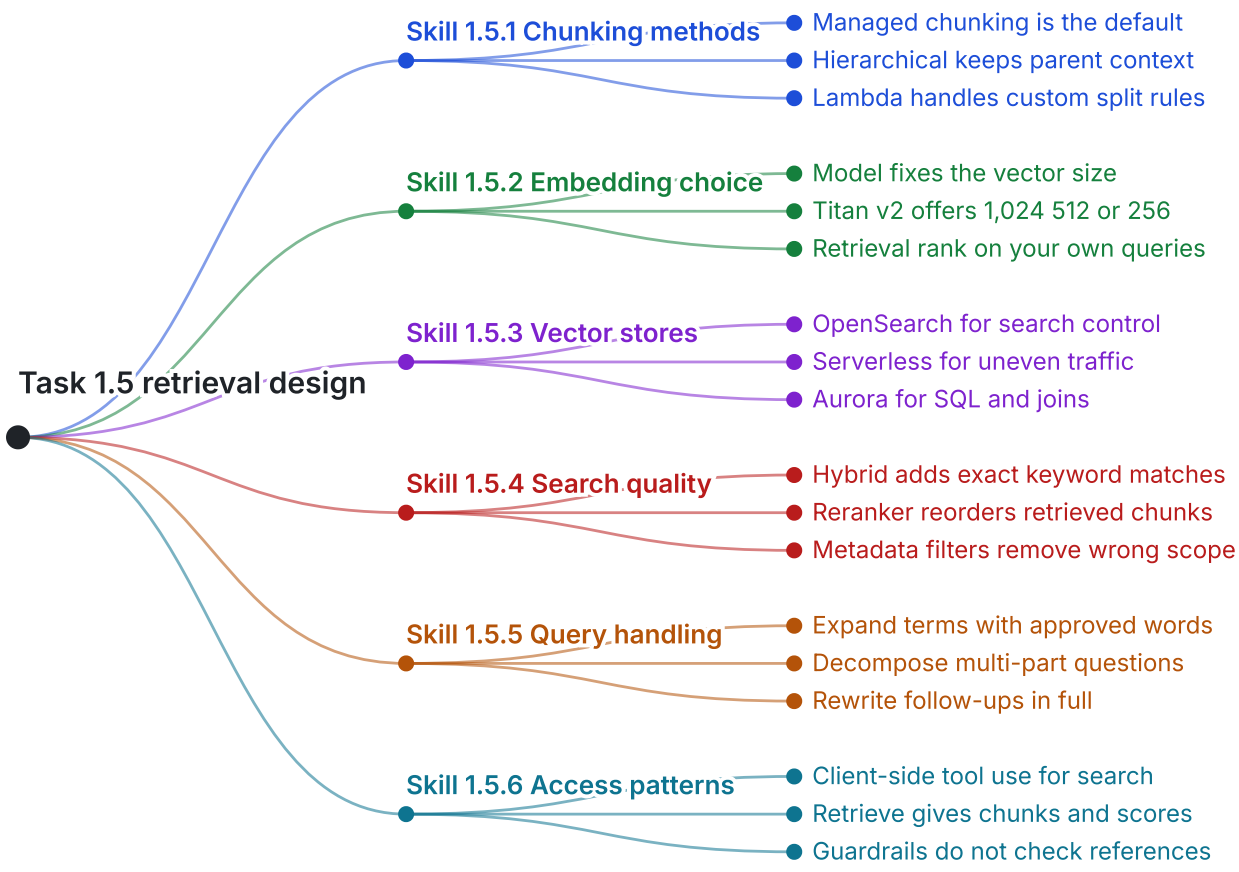
Memory sheet: Task 1.5 Design retrieval mechanisms for FM augmentation

Domain 1, weight 31 percent. Read this sheet, then answer the flash questions aloud.



Deciding rules

When the stem says	Think of	Not when
"fixed size" or "overlap"	Fixed-size chunking with overlap	The section must stay whole, since fixed size can split a long section
"parent section" or "surrounding context"	Hierarchical chunking	The answer sits in one short paragraph, so parent text adds tokens only
"exact code", "SKU", or "rare name"	Hybrid search	The right chunk ranks too low, which needs a reranker instead
"right result, wrong order"	Reranker	The right chunk never came back from retrieval, so widen retrieval first
"only for one region" or "only approved"	Metadata filter built on the server from a signed token	A browser field sets the scope, since the client can change it
"multi-part question"	Query decomposition with one retrieval per part and a merge	The question has a single topic, so extra calls add latency with no gain
"tool the model can request"	Client-side tool use that calls Retrieve	One fixed retrieval step is enough, so an agent loop adds cost

Etch lines

1. Find the constraint first, since it usually decides the store and chunk shape.
2. Managed chunking is the first option to test before custom code.
3. Hybrid search widens the match, but it does not reorder them by relevance.
4. Titan v2 gives 1,024 dimensions by default, with 512 or 256 as options.
5. Use Retrieve when your code must see chunks before the model runs.
6. A guardrail does not inspect retrieved references, so check them in code.

Flash questions

- Q: A support bot answers from Japanese manuals, while customers write in English. What is the main concern? A: Matches will be weaker, so pick a model that fits the corpus language.
- Q: Reranking is enabled, but the answer chunk is missing from the candidates. Which stage is at fault? A: Retrieval, since a reranker only reorders what retrieval returned.
- Q: Results must stay within one region. Which technique removes out-of-scope chunks? A: A metadata filter applied on the server side.
- Q: An app must inspect chunks before the model reads them. Which operation fits? A: Retrieve, called from the app code or from a client-side tool.
- Q: A team trusts its guardrail to screen the passages that retrieval returns. Is that correct? A: No, guardrails screen the input and the generated response, not retrieved references.
- Q: After a model upgrade, vectors arrive at 1,024 dimensions while the index expects 512. What is required? A: Rebuild the index at the model size, since the two must match.

- Q: A design needs a named vector store, so which knowledge base type cannot be used? A: The managed type, because it hides the store choice.
- Q: A parent chunk size grows large while the store is an S3 vector bucket. What goes wrong? A: Hierarchical chunking is not recommended there, and parent metadata can exceed its size limit.

Source: Network and account placement, Skills 1.5.1 to 1.5.6, X versus Y confusable pairs, Read a long question, Distractor patterns, and Drill. AWS docs and the study guide sections above, no practice-test text.